# Computer Vision Applications

Computer vision is a branch of artificial intelligence that trains machines to interpret and understand visual information from the world, such as images and videos. The field draws on techniques from image processing, geometry, and increasingly, deep learning. A foundational task in computer vision is image classification, where a model assigns a label to an entire image based on its content. Object detection extends this idea by identifying multiple objects within an image and drawing bounding boxes around each one, while image segmentation goes further by classifying every individual pixel in an image. Convolutional neural networks have become the standard tool for these tasks, learning hierarchical features that progress from simple edges and textures in early layers to complex shapes and objects in deeper layers. Popular architectures such as ResNet, YOLO, and Vision Transformers have pushed the boundaries of accuracy and speed in visual recognition tasks. Real-world applications of computer vision are extensive, including facial recognition systems used for security, medical imaging tools that detect tumors or fractures from X-rays and MRI scans, and autonomous vehicles that must interpret their surroundings in real time to navigate safely. Quality control systems in manufacturing use computer vision to detect defects on production lines far faster and more consistently than human inspectors. As with other AI applications, computer vision systems can inherit biases present in their training data, and ensuring fairness and robustness across diverse populations and conditions remains an active area of research.